Algebraic Expression

An **algebraic expression** is a combination of constants and variables combined together with the help of the four fundamental signs.

Examples of algebraic expression are

$$x^3 - 4x^2 + 8x - 1,\ 4xy^2 + 3x^2y - \frac{5}{4}xy + 9,\ 5x^2 - 7x + 6$$

Coefficients

Any part of a term that is multiplied by the variable of the term is called the coefficient of the remaining term.

For example,

$x^2 + 5x - 24$ is an algebraic expression containing three terms. The variable of this expression is x, coefficient of x^2 is 1, the coefficient of x is 5 and the constant is –24 (not 24).

Activity-1

Write the Variable, Coefficient and Constant in the given algebraic expression

Expression	$x + 7$	$3y - 2$	$5x^2$	$2xy + 11$	$-\frac{1}{2}p + 7$	$-8 + 3a$
Variable	x			x, y		
Coefficient	1				$-\frac{1}{2}$	
Constant	7					–8

3.2 Polynomials

A polynomial is an arithmetic expression consisting of variables and constants that involves four fundamental arithmetic operations and non-negative integer exponents of variables.

Polynomial in One Variable

An algebraic expression of the form $p(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_2x^2 + a_1x + a_0$ is called Polynomial in one variable x of degree 'n' where $a_0, a_1, a_2, \ldots a_n$ are constants ($a_n \neq 0$) and n is a whole number.

In general polynomials are denoted by $f(x), g(x), p(t), q(z)$ and $r(x)$ and so on.

Note

The coefficient of variables in the algebraic expression may have any real numbers, where as the powers of variables in polynomial must have only non-negative integral powers that is, only whole numbers. Recall that $a^0 = 1$ for all a.

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